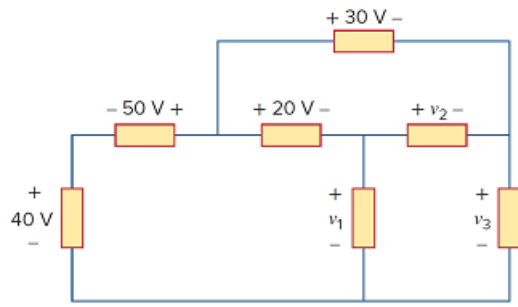


Problem

In the circuit in Fig. 2.76, obtain v_1 , v_2 , and v_3 .

Figure 2.76:



Step-by-step solution

Step 1 of 5

The circuit is shown in Figure 1.

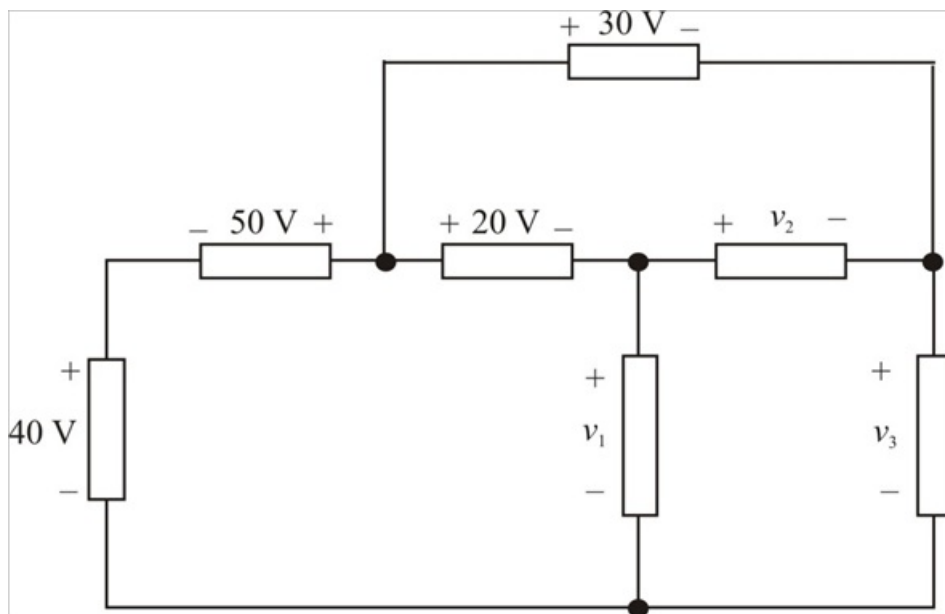


Figure 1

Step 2 of 5

Apply loops to the circuit as shown in Figure 2.

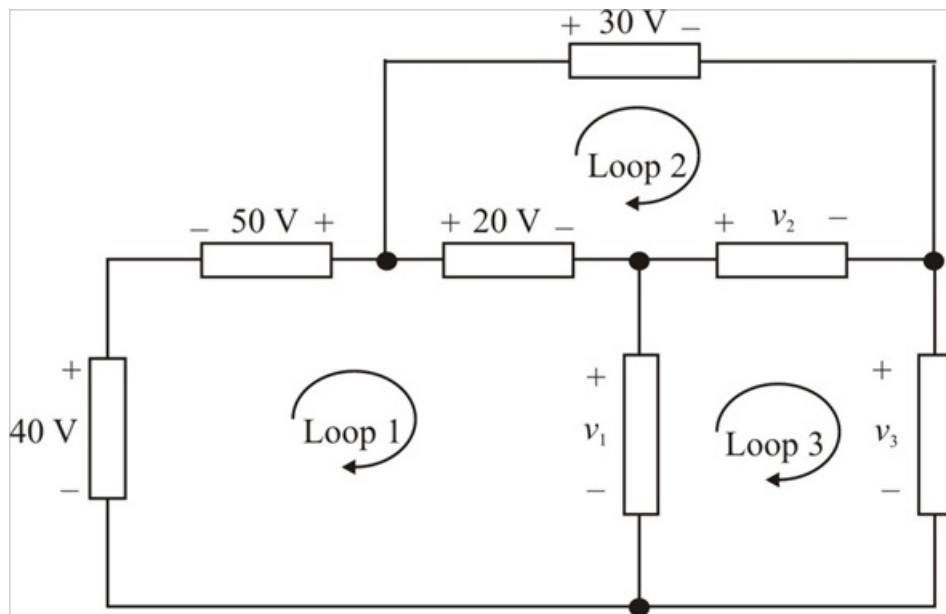


Figure 2

Step 3 of 5

Apply Kirchhoff's voltage law, the algebraic sum of all voltages around a closed path (or loop) is zero. Calculate the voltage v_1 .

$$(-40) + (-50) + 20 + v_1 = 0$$

$$v_1 = 70 \text{ V}$$

Therefore, the voltage v_1 is 70 V.

Step 4 of 5

Apply Kirchhoff's voltage law, calculate the voltage v_2 .

$$30 - v_2 - 20 = 0$$

$$v_2 = 10 \text{ V}$$

Therefore, the voltage v_2 is 10 V.

Step 5 of 5

Apply Kirchhoff's voltage law, calculate the voltage v_3 .

$$-v_1 + v_2 + v_3 = 0$$

Substitute 70 for v_1 and 10 for v_2 .

$$-70 + 10 + v_3 = 0 \quad -v_1 + v_2 + v_3 = 0$$

$$v_3 = 60 \text{ V}$$

Therefore, the voltage v_3 is 60 V.